# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The thickness of a hair is about 0.1 mm. In metres it is: 1

**(a)** 0.0001 m **(b)** 0.01 m **(c)** 0.1 m **(d)** 0.001 m

**2.** Which is the best estimate of the length of a school bus? 1

**(a)** 10 km **(b)** 10 m **(c)** 10 cm **(d)** 10 mm

**3.** The Earth rotates once on its axis in about: 1

**(a)** 12 hours **(b)** 365 days **(c)** 27 days **(d)** 24 hours

**4.** A measuring tape has marks in cm on one side and inches on the other. This is: 1

**(a)** because the tape is faulty **(b)** to measure time

**(c)** to show lengths in two different units **(d)** to make the tape flexible

**5.** The smallest division on a ruler marked in millimetres is: 1

**(a)** 1 mm **(b)** 0.1 mm **(c)** 1 cm **(d)** 1 m

**6.** The Earth revolves around the Sun in about: 1

**(a)** 30 days **(b)** 27 days **(c)** 365 days **(d)** 24 hours

**7.** Walking 5 steps each of 0.6 m, the distance walked is: 1

**(a)** 0.6 m **(b)** 5 m **(c)** 3 m **(d)** 30 m

**8.** A metre scale has the zero mark worn. A reading from 1 cm to 11.6 cm means length: 1

**(a)** 11.6 cm **(b)** 10.6 cm **(c)** 12.6 cm **(d)** 1 cm

**9.** **Assertion (A):** A car moving along a straight road shows circular motion. 1

**Reason (R):** Circular motion means moving along a circle.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** To read a scale correctly, the eye must be placed directly above the mark being read. 1

**Reason (R):** Reading from a slant gives an error called parallax.

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** A scooter wheel of circumference 1.5 m turns 200 times. What distance is covered? 2

**12.** Give an example where an object is both at rest and in motion at the same time. 2

**13.** How would you measure the length of a river on a school map? 2

**14.** Why does the Sun appear to move from east to west? 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** A runner covers 400 m in each lap. How many laps for 5 km? 3

**16.** Why should a measuring tape be held straight and tight while measuring a long length? 3

**17.** A path has three straight sections of 2.5 m, 450 cm and 0.0036 km. Find total length in metres. 3

OR

Two students measure the length of the same desk and get 1.28 m and 128.4 cm. Explain whether their results agree. 3

**18.** Explain why the estimate of length by eye is less reliable than measurement. 3

OR

Why does a spring bounce periodically? 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** At a fair in Mysuru, Neha saw a toy train moving along a straight track, a giant wheel turning round and round, and a swing moving to and fro. A butterfly was flying here and there around the stalls. 5

(i) What kind of motion does the toy train show?

(ii) Which of these is a periodic motion?

(iii) Which one shows random motion?

(iv) The giant wheel completes one turn in 2 minutes. Is its motion periodic? Explain.

**20.** Explain with examples how the reference point decides whether an object is at rest or moving. 5

OR

A girl walks twice around a square park of side 120 m. Find the total distance she walks. 5

## END OF QUESTION PAPER — Total: 40 Marks